

Gadolinium Adsorption by Functionalized Mesoporous Silicas (MCM-41/SBA-15): A Proposal for Volume and Dose Reduction in Radioactive Waste

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INTRODUCTION

The increasing use of gadolinium-based contrast agents (GBCAs) in medical imaging has led to the accumulation of gadolinium (Gd) in aquatic environments, where the breakdown of complexes can release toxic Gd³⁺ ions.

Mesoporous silica materials, such as MCM-41 and SBA-15, with high surface area, tunable porosity, and functionalizable surfaces, are promising for selective metal ion adsorption. This study evaluates MCM-41 and SBA-15 functionalized with di(2-ethylhexyl)phosphoric acid (D2EHPA) for Gd³⁺ removal from water, exploring synthesis routes, physicochemical characterization, and adsorption tests, aiming to provide effective strategies to mitigate gadolinium contamination.

Mesoporous materials have low activity in their pure form due to the lack of active sites. Incorporating chemical groups into their porous walls expands their applications in catalysis, separation, molecular sieves, and adsorption. A common modification strategy for materials like MCM-41 and SBA-15 is the immobilization of metals or metal complexes, achieved by introducing organic groups through reaction with surface silanol groups, which preserves the material's ordered structure and ensures high hydrothermal stability.

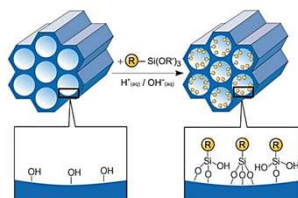


Figure 1: Scheme of MCM-41 modification with organic groups.
Source: Hoffmann et al., 2006.

MATERIALS AND METHODS

• Synthesis of MCM-41

MCM-41 was synthesized under three conditions to evaluate the effect of the synthesis route on material properties. In the standard method (MCM-41(1-A)), CTABr in NaOH was mixed with TEOS, stirred, aged, and hydrothermally treated at 100 °C for 72 h. The reflux method (MCM-41(1-R)) heated the same gel under reflux at 100 °C for 72 h, while the third route (MCM-41(1-T)) used high-speed mechanical stirring before aging and heat treatment.

• Synthesis of SBA-15

SBA-15 was synthesized under acidic conditions using two methods. Pluronic P123 in HCl was mixed with TEOS and stirred for 24 h for hydrolysis and condensation. In the autoclave route (SBA-15(A)), the mixture underwent hydrothermal treatment at 100 °C for 24 h, while in the reflux route (SBA-15(R)) it was heated under reflux at 100 °C for the same time. Solids were filtered, washed, dried at 60 °C, and calcined at 500 °C. Functionalization was done by wet impregnation with D2EHPA: 1 g of silica was stirred with 10 mL of 30% D2EHPA in kerosene for 1 h, solvent removed under vacuum, and the material dried at 60 °C for 24 h.

RESULTS

• Gadolinium Adsorption Tests

FTIR spectra were recorded on KBr pellets from 4000–400 cm⁻¹. Calcined materials showed characteristic silica vibrations, while D2EHPA-functionalized samples displayed new bands at 1225 cm⁻¹ (P=O) and 1040 cm⁻¹ (P–OH), confirming phosphate groups on the surface. Broad bands at 3450 cm⁻¹ and 1630 cm⁻¹ corresponded to surface hydroxyls and adsorbed water.

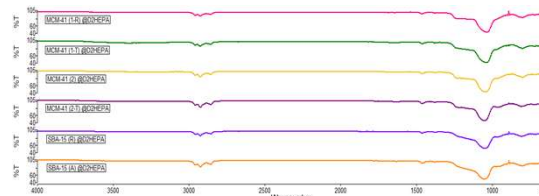


Figure 2: FTIR spectra after functionalization of MCM-41 and SBA-15 with D2EHPA.

Surface area, pore volume, and pore diameter were measured by nitrogen adsorption-desorption at 77 K using BET and BJH methods. MCM-41 samples showed high surface areas (890–1100 m²·g⁻¹) with narrow pores (2.5–3.5 nm), while SBA-15 had lower surface areas (500–750 m²·g⁻¹) but larger pores (5–9 nm). MCM-41(2) was selected for adsorption tests due to its highest surface area and potential for enhanced Gd³⁺ retention.

Table 1: Gd³⁺ Sorption as a Function of Contact Time Between Resin and Solution

Contact Time (min)	Initial Gd ³⁺ Concentration (µg·L ⁻¹)	Gd ³⁺ Concentration After Sorption (µg·L ⁻¹)	Gd ³⁺ Retention (%)
5	25.8 ± 4.0	7 ± 6	67 ± 30
10	35.6 ± 7.3	12 ± 6	66 ± 11
15	43.5 ± 2.6	0.8 ± 0.2	98.0 ± 0.4
20	35 ± 10	2 ± 0.8	93.1 ± 4.4
25	11.7 ± 3.3	0.62 ± 0.05	94.3 ± 1.1
30	60.0 ± 4.4	2 ± 0.5	96.8 ± 0.4
35	59 ± 15	1 ± 0.2	97.7 ± 0.3
40	21 ± 17	0.40 ± 0.09	95.9 ± 2.7
45	36.3 ± 3.9	0.4 ± 0.15	98.8 ± 0.5
50	31.4 ± 1.4	0.21 ± 0.03	99.3 ± 0.1
55	43.0 ± 4.2	0.4 ± 0.2	98.9 ± 0.6
60	39.6 ± 1.7	1 ± 1	97.0 ± 2.6

CONCLUSIONS

MCM-41 and SBA-15 functionalized with D2EHPA efficiently removed Gd³⁺ from aqueous solutions and effluents, with MCM-41(2) showing the best performance. Under optimized conditions—pH 3, 15 min contact, and 50 mg adsorbent—it achieved over 99% gadolinium retention. FTIR and BET analyses confirmed successful functionalization and the impact of synthesis on textural properties. The rapid, high-capacity adsorption demonstrates the material's potential for environmental remediation and radioactive waste treatment, offering a safe and effective method for gadolinium immobilization.

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